Pranav Maturi

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Education

Georgia Institute of Technology — Atlanta, GA
Bachelor of Science in Computer Engineering | Institute GPA: 4.0 — *Expected May 2030*

- **Relevant Coursework**: Object-Oriented Programming, Algorithms, Data Structures, Memory Management, Linear Algebra, Multivariable Calculus

Technical Skills & Certifications

Languages: Python, C, Java, Swift, SystemVerilog, SQL, HTML5, CSS3, JavaScript
Tools & Frameworks: Cadence SimVision, FastX, Flask, SQLAlchemy, Bootstrap 5, Git, SQLite, Cisco Packet Tracer
Certifications: Harvard CS50x (2026), Tech Specialist in Software Development (2025), MOS (2022)

Projects

32-bit RISC-V Single-Cycle Processor (Silicon Jackets) | *SystemVerilog, Cadence* — Sep 2026

- Architected a functional 32-bit single-cycle RV32I RISC-V processor in SystemVerilog, engineering core logic modules including the controller, register file, and immediate generator utilizing combinational logic.
- Configured enterprise CAD toolchains on remote ECE research servers via C-shell, and debugged electrical voltage waveforms utilizing Cadence SimVision over a FastX remote desktop.
- Authored comprehensive architectural documentation detailing hardware behaviors, memory latency, and addressing schemes to successfully complete the Silicon Jackets digital design onboarding.

Smart Microblog | *Python, Flask, SQLAlchemy, TextBlob, AJAX, JS, CSS* — Aug 2026

- Engineered a full-stack microblogging platform featuring real-time AI sentiment analysis via TextBlob, dynamically color-coding user feeds based on mood classification.
- Implemented asynchronous AJAX polling for seamless timeline updates without page reloads, and designed a modern glassmorphism UI utilizing IntersectionObserver scroll-reveal animations.
- Integrated Flask-SQLAlchemy for robust database management and utilized Werkzeug for secure password hashing.

Community-Driven Appointment Scheduler | *Python, Flask, SQLite3, Bootstrap 5* — Aug 2026

- Architected a scalable web application to centralize peer-to-peer tutoring logistics and educational resource sharing.
- Engineered a normalized relational database utilizing complex SQL JOIN queries to dynamically render dual-view user dashboards for accounts acting simultaneously as students and instructors.

Accessibility iOS App & Cybersecurity Competitor | *Swift, Cisco Packet Tracer* — 2025 – 2026

- Developing a specialized iOS application providing customized entertainment interfaces for individuals with disabilities.
- Competed in national-level cybersecurity competitions, successfully placing in the platinum league with team members.

Experience

Georgia Tech VIP Program (EcoCAR) — Atlanta, GA
Undergraduate Researcher – Connected and Automated Vehicles (CAV) — *Aug 2026 – Present*

- Collaborating on developing a Connected and Automated Vehicle (CAV) software stack, integrating advanced sensing arrays (LiDAR, radar, vision) to optimize autonomous perception and vehicle safety.
- Executing software-in-the-loop (SIL) simulations and rigorous testing protocols to validate dynamic control algorithms, ensuring robust performance and real-world system compliance.

Schoolhouse — Remote
Dialogues Club President & Tutor — *2021 – 2026*

- Developed curriculum and hosted an international-level forum for 100+ students, boosting engagement by 50%.
- Tutored students for the SAT math section, contributing to measurable score improvements over 40+ volunteer hours.

Mathnasium — Cumming, GA
Lead Instructor — *June 2025 – May 2026*

- Instructed students and guided coursework to achieve academic goals utilizing the specialized Mathnasium Method.
- Documented performance results, updated customized learning plans, and communicated progress metrics with parents.

Adaptive Dance for Disabled Association (A.D.D.A) — Cumming, GA
Co-President — *2021 – 2026*

- Directed a weekly dance club for special education students, independently managing logistics and curriculum.
- Choreographed adaptive routines and provided specialized instruction to approximately 10 students per session.

Awards & Honors

Awards: ACSL Finals Qualifier, HackerRank Senior Division Finalist, FBLA Intro to Programming Competitor
Honors: AP Scholar with Distinction, Georgia Certificate of Merit, National Honors Society